

# Interdiffusion of Cr, Mo, and W in Iron

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The diffusion rates of chromium, molybdenum, and tungsten in  $\alpha$ - and  $\gamma$ -iron have been determined by electron-probe microanalysis. Alloy concentrations up to 28.3 at.-% Cr, 5.05 at.-% Mo, 2.9 at.-% W have been investigated over the range 675 to 1485° C. At any temperature in either ferrite or austenite,  $\bar{D}_{Mo} > \bar{D}_{Cr} > \bar{D}_W$ . Interdiffusion in each alloy system was found to be independent of concentration for the composition ranges studied. Data from the present investigation have been combined with existing reliable data so that diffusion in these alloy systems may be described by the following values of frequency factor ( $D_0$ ) and activation energy ( $Q$ ), with errors given at the 95% confidence limits:

|            | $D_0$ ,<br>mm <sup>2</sup> /s | $Q$ ,<br>kJ/mol | $D_0$ ,<br>cm <sup>2</sup> /s | $Q$ ,<br>kcal/mol |
|------------|-------------------------------|-----------------|-------------------------------|-------------------|
| Alpha-Iron |                               |                 |                               |                   |
| Cr         | 240 + 14<br>- 13              | 239.8 ± 4.2     | 2.40 + 0.14<br>- 0.13         | 57.3 ± 1.0        |
| Mo         | 78.5 + 6.4<br>- 5.9           | 225.5 ± 4.6     | 0.785 + 0.064<br>- 0.059      | 53.9 ± 1.1        |
| W          | 157 + 18<br>- 16              | 243.5 ± 7.1     | 1.6 ± 0.2                     | 58.2 ± 1.7        |
| Gamma-Iron |                               |                 |                               |                   |
| Cr         | 6.27 + 0.46<br>- 0.42         | 252.3 ± 18.4    | 0.063 + 0.005<br>- 0.004      | 60.3 ± 4.4        |
| Mo         | 3.60 + 0.40<br>- 0.36         | 239.8 ± 26.4    | 0.036 + 0.005<br>- 0.004      | 57.3 ± 6.3        |
| W          | 13.0 + 16.0<br>- 7.0          | 267.4 ± 10.8    | 0.13 + 0.16<br>- 0.07         | 63.9 ± 2.6        |

For diffusion in ferrite, the tracer (literature) and interdiffusion (literature and present work) coefficients for Cr, Mo, and W generally show close agreement. However, the available data for diffusion in austenite suggest that for Cr and W interdiffusion is significantly slower than tracer diffusion. No firm conclusions as to the explanation of this discrepancy are given, but the possibility of a systematic error in the tracer measurements is discussed.

A knowledge of the diffusion rates for various alloying elements in iron is desirable because of the technological importance of iron-base alloys. The three elements studied in this investigation (Cr, Mo, W) come from Group VI of the periodic table. Existing data for these systems are limited and there are large discrepancies between reported measurements. The purpose of this investigation was to produce reliable diffusion data for each of the three systems studied.

## Experimental Procedure

High-purity iron (2082 BVI) was produced by vacuum-melting high-purity Japanese electrolytic iron in an alumina crucible. This was extruded and hot swaged at 1000° C to produce 7 mm-dia. rod. The analyses of this iron, of similar iron from a different source (951 BV), and of the other alloys used are given in Table I.

For the iron-chromium system, a range of high-purity, vacuum-melted alloys was available.<sup>1</sup> These had been prepared using the same iron stock as 2082. For the other systems studied, the pure iron ingot (2082 BVI) was the base metal and appropriate additions of pure 'Murex' molybdenum powder (99.8 + %) and pure 'Murex' tungsten rod (99.8 + %) were used to produce vacuum-melted 1 kg ingots, 25 mm in diameter. Each alloy was hot rolled and hot swaged at 1100° C to produce 7 mm-dia. rod. Any oxide scale formed in this process was carefully machined off. The homogeneity of the major alloy constituent from each cast was checked, after swaging, by chemical analysis and EPMA (see Table I).

The 7 mm-dia. alloy rods were cut into 12.5 mm lengths and two parallel flat faces were produced along the length of each cylindrical piece by cold rolling. Each length was then mounted in a plastic holder and both flat faces ground on 600 grade SiC paper. The alloy rods were joined to pure iron (Fig. 1) by resistance-welding in an atmosphere of flowing argon. After welding, the specimen was sectioned into two pieces and the ends of each were examined microscopically to ensure that the interface was sound and contained no holes or oxide inclusions. Specimens produced and examined in this way were required to have a grain size of ~ 400  $\mu$ m, which would normally exceed twice the length of an expected diffusion profile.

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TABLE I  
Alloy Compositions (wt.-%)

| Element      | Cast Number |            |            |            |            |            |            |           |           |            |           |           |           |
|--------------|-------------|------------|------------|------------|------------|------------|------------|-----------|-----------|------------|-----------|-----------|-----------|
|              | 2082<br>BV1 | 2139<br>RV | 1291<br>RV | 1576<br>RV | 1577<br>RV | 1294<br>RV | 2215<br>RV | Ar<br>114 | Ar<br>116 | 2214<br>RV | Ar<br>115 | Ar<br>117 | 951<br>BV |
| Cr           | <0.02       | 1.15       | 1.38       | 10.7       | 13.3       | 26.9       | <0.02      | <0.02     | <0.02     | <0.02      | <0.02     | <0.02     | <0.02     |
| Mo           | <0.02       | <0.02      | <0.02      | <0.02      | <0.02      | <0.02      | 0.95       | 4.54      | 9.20      | <0.02      | <0.02     | <0.02     | <0.02     |
| W            | <0.02       | <0.02      | <0.02      | <0.02      | <0.02      | <0.02      | <0.02      | <0.02     | <0.02     | 1.02       | 4.45      | 10.45     | <0.02     |
| C            | 0.002       | 0.005      | 0.006      | 0.005      | 0.004      | 0.006      | 0.002      | 0.002     | 0.003     | 0.002      | 0.003     | 0.005     | 0.018     |
| Si           | <0.02       | 0.02       | 0.02       | 0.03       | <0.02      | 0.04       | <0.02      | <0.02     | <0.02     | <0.02      | <0.02     | <0.02     | <0.02     |
| Al           | 0.005       | 0.005      | <0.02      | 0.012      | 0.009      | 0.015      | 0.005      | <0.02     | <0.02     | 0.005      | <0.02     | <0.02     | <0.02     |
| O            | 0.013       |            |            |            |            | 0.008      |            |           |           |            |           |           |           |
| N            | 0.005       |            | 0.02       |            |            | 0.004      |            |           |           |            |           |           | 0.002     |
| Homogeneity* | 0.1         | 0.1        | 0.1        | 0.2        | 0.1        | 0.1        | 0.02       | 0.1       | 0.1       | 0.02       | 0.01      | 0.1       |           |

Ni, V, Nb, Ti, Co, Mn, S, and P all <0.02 wt.-%

\* Homogeneity figures indicate the variation of major alloying constituents from the top to the bottom of the ingot.

Experiments were carried out on diffusion couples which could be of the same phase throughout or could change crystal structure due to change in composition at the diffusion temperature. All heat-treatments were carried out in furnaces with long-term stability better than  $\pm 2$  degC and with a furnace control of  $\sim \pm \frac{1}{2}$  degC. Temperatures were measured by means of Pt/Pt-13% Rh thermocouples sheathed in recrystallized alumina. The thermocouples were checked periodically against a known standard thermocouple, which was carefully preserved. Agreement was always within  $\pm \frac{1}{2}$  degC at 1100° C.

Diffusion couples were encapsulated in silica tubes 30 mm long  $\times$  10 mm dia. under a partial pressure of pure argon. After each diffusion anneal, specimens were rapidly removed from the furnace by means of a nichrome or platinum wire attached to the capsule and quenched immediately into water. Quenching times were estimated as being not more than 5 s.

After quenching, each diffusion couple was sectioned perpendicularly to the welded interface and the central portion polished, in a plane parallel to the diffusion direction, to  $\frac{1}{4}$   $\mu$ m diamond finish. After a light etch, grain-boundary-free areas in excess of 400  $\mu$ m were marked out at the diffusion interface by means of microhardness impressions. The specimen was then repolished to  $\frac{1}{4}$   $\mu$ m.

Microanalysis was carried out using a Cambridge Scientific Instrument Co. 'Microscan V'. A correction procedure based on Duncumb's ZAF correction<sup>2,3</sup> was applied. Concentrations of standard alloys measured in this way gave good agreement with the known chemical compositions. Concentration profiles were measured by use of the automatic step-scan facility on the instrument. For welded specimens that had not been heat-treated, the measured profiles were found to be extremely small ( $< 5$   $\mu$ m). Since the diffusion profiles were always  $> 50$   $\mu$ m, blank corrections were not necessary.

#### Treatment of Results

The Boltzmann-Matano analysis is a practicable method that allows the evaluation of interdiffusion coefficients as a function of concentration in single-phase and in multilayer diffusion couples. Let  $C(x, t)$  be the solute concentration at time ( $t$ ) and distance ( $x$ ) for a semi-infinite diffusion couple. Initially, each side of the couple is at a uniform composition of  $C = C_0$  and  $C = C_s$ , respectively, these compositions

remaining constant at large distances from the original interface throughout the diffusion anneal. Where the original interface is initially a phase boundary (which at the annealing temperature divides two different structures), it is assumed that boundary movement  $\propto \sqrt{t}$ .

A solution to Fick's second equation, where the chemical interdiffusion coefficient  $D(c)$  is a function of composition, leads to a definition of the Matano interface as the zero of the  $x$  co-ordinate, where

$$\int_{C_0}^{C_s} x dc = 0$$

and a value of the interdiffusion coefficient

$$\bar{D}_{C1} = -\frac{1}{2t} \left( \frac{dx}{dc} \right) \int_{C_0}^{C_1} x dc$$

The value of  $\bar{D}_{C1}$  at  $C = C_1$  is calculated from the reciprocal of the slope of the concentration/distance curve at  $C = C_1$  and the area under the curve up to that point. For the concentrations studied no correction for volume change was necessary. A computer programme was used to take the tape output from the microanalyser, convert the observed X-ray counts into concentrations, and perform the above analysis.

The Grube<sup>6</sup> analysis was applied to evaluate the diffusion coefficient ( $\bar{D}$ ), when it was found not to vary over the experimental range of composition in the diffusion couple. The appropriate solution for constant  $\bar{D}$  is

$$\frac{C(x, t) - C_0}{C_s - C_0} = \frac{1}{2} \left( 1 - \operatorname{erf} \left( \frac{x}{2 \sqrt{\bar{D}t}} \right) \right)$$

A plot of  $(C - C_0)/(C_s - C_0)$  on a probability scale will result in a straight line, provided  $\bar{D}$  is constant, with a slope proportional to  $(\bar{D}t)^{-1/2}$ . Deviations from linearity over the central regions of this plot indicate the extent to which  $\bar{D}$  is a function of composition. This analysis can be conveniently used to evaluate  $\bar{D}$  as  $C \rightarrow 0$ , and for the results of the present work was found to be generally applicable.

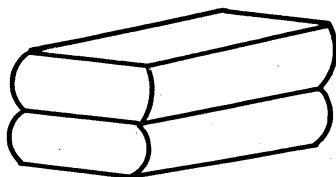


Fig. 1 Diffusion couple,  $\sim 8$  mm wide  $\times$  10 mm high  $\times$  12.5 mm long.

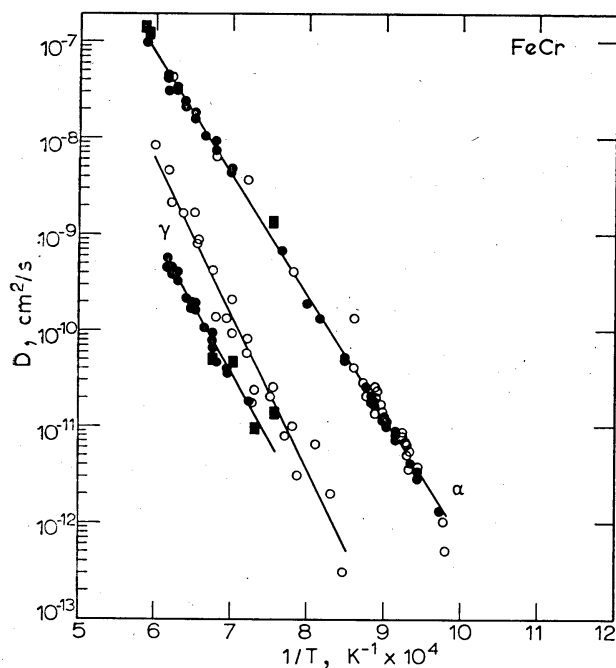


Fig. 2 Interdiffusion and tracer diffusion for FeCr. ● Present investigation; ■ interdiffusion data (Refs. 7-9); ○ tracer diffusion data (Refs. 8, 15-18).

### Results

The results of the present investigation are presented in Figs. 2-4) and are summarized by the values of  $D_0$  and  $Q$  shown in Table II. All values of interdiffusion coefficients were found to be independent of composition over the experimental ranges. The results of other workers, for those measurements considered to be reliable, are as follows.

Heumann and Bohmer<sup>7</sup> found interdiffusion coefficients in ferrite in iron-chromium alloys containing up to 24 at.-% Cr were independent of composition between 10-24 at.-% Cr. Values of  $D_0$  and  $Q$  derived from these data are not considered to be reliable since only a few measurements were taken. However, the absolute values of interdiffusion coefficient shown in Fig. 2 agree with the measurements of the present investigation. Below 10 at.-% Cr, a slight concentration-dependence was noted,<sup>7</sup>  $\bar{D}_\alpha$  increasing by  $\sim 30\%$  over the range 10  $\rightarrow$  0 at.-% Cr. However, large deviations were reported at each end of the concentration profile and the observed dependence on concentration is thought to result from inaccuracies in the analysis at the tail of the concentration profile.

Interdiffusion in austenite was also studied<sup>7</sup> and found to be independent of chromium concentration. Only 3 measurements were made in the range 950-1150° C. It has been shown<sup>8</sup> that values for interdiffusion in austenite in the lower part of

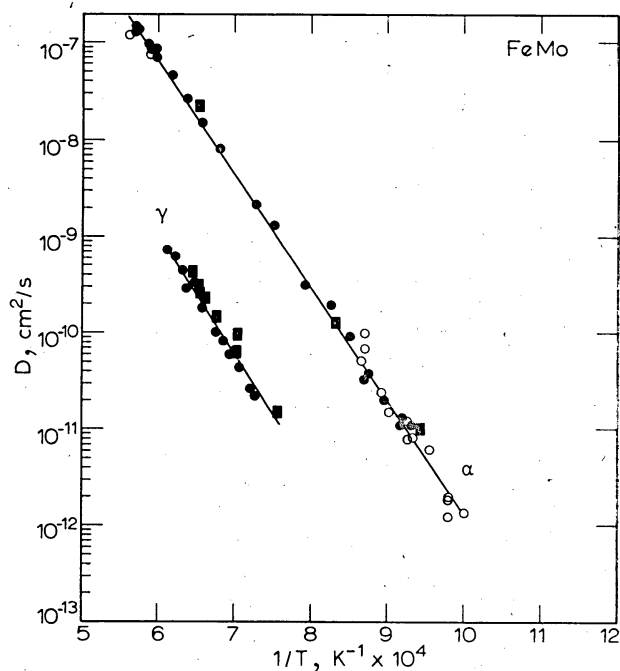


Fig. 3 Interdiffusion and tracer diffusion for FeMo. ● Present investigation; ■ interdiffusion data (Refs. 10-12); ○ tracer diffusion data (Refs. 19, 20).

TABLE II  
Interdiffusion Data Derived from Measurements of the Present Investigation\*

|    | Alpha-Iron                 |                 | Gamma-Iron                 |                  |
|----|----------------------------|-----------------|----------------------------|------------------|
|    | $D_0$ , mm <sup>2</sup> /s | $Q$ , kJ/mol    | $D_0$ , mm <sup>2</sup> /s | $Q$ , kJ/mol     |
| Cr | $220 + 50$<br>$- 40$       | $239.7 \pm 1.6$ | $10 + 10$<br>$- 5$         | $258.6 \pm 9.2$  |
| Mo | $83 + 25$<br>$- 19$        | $225.5 \pm 3.4$ | $11 + 33$<br>$- 9$         | $255.2 \pm 15.0$ |
| W  | $110 + 60$<br>$- 40$       | $238.9 \pm 5.8$ | $13 + 16$<br>$- 7$         | $267.4 \pm 10.8$ |

\* Errors are expressed to 95% confidence limits.

this temperature range are increasingly susceptible to grain-boundary effects, and so only the more reliable measurements are shown in Fig. 2. Further measurements<sup>9</sup> of interdiffusion in austenite using alloys containing up to 7.5 at.-% Cr have been reported over the range 900-1200° C. It was realized that a grain-boundary contribution probably influenced some of these and those results between 900 and 1050° C are omitted from Fig. 2. It can be seen that there is good agreement between existing measurements of interdiffusion coefficients above 1050° C.

In the iron-molybdenum system two experiments by Ham,<sup>10</sup> one at 931° C and the other at 1260° C, were used to determine interdiffusion coefficients in ferrite between two alloys with 1.9 and 3.6 at.-% Mo, respectively. The interdiffusion coefficients were found to be independent of concentration over this composition range and are shown in Fig. 3. In addition, interdiffusion in austenite was measured<sup>10</sup> over the range 1151-1263° C for alloys containing up to 1.4

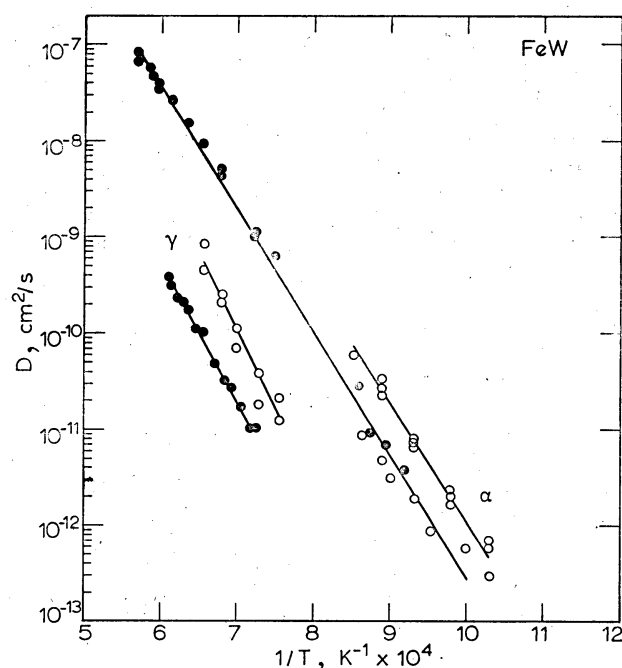


Fig. 4 Interdiffusion and tracer diffusion for FeW. ● Present investigation; ○ tracer diffusion data (Refs. 8, 20–22).

at.-% Mo. No concentration-dependence was observed and these results are also shown in Fig. 3. Pivot<sup>11</sup> *et al.* studied interdiffusion in ferrite over the range 790–1185°C. However, the presence of austenite in measurements above 910°C was neglected in the resultant analysis. This was not justified in these measurements and only those results between 790 and 910°C have been included in Fig. 3. No concentration-dependence was observed. Krishtal and Mokrov<sup>12</sup> carried out four measurements of interdiffusion coefficients in austenite over the range 1050–1280°C on an iron-molybdenum alloy containing 2 at.-% Mo. No evidence of a concentration-dependence was revealed. The reported values of interdiffusion coefficients are shown in Fig. 3. It can be seen that these measurements agree with those of the present investigation and those of Ham.<sup>10</sup>

For the iron-tungsten system the results of the present investigation are the first to be published for interdiffusion in ferrite and austenite.

### Discussion

The more reliable measurements reported above<sup>7,9–12</sup> agree closely with the results of the present investigation as shown in Figs. 2–4. For interdiffusion in each of the three systems there is no evidence to suggest a dependence on concentration for alloys containing up to 28.3 at.-% Cr, 5.05 at.-% Mo, or 2.9 at.-% W.

Interdiffusion coefficients are related to tracer diffusion coefficients<sup>13</sup> by the equation

$$\bar{D}_{AB} = (N_A D_B^* + N_B D_A^*) \left( 1 + \left( 1 + \frac{\partial \ln \gamma_B}{\partial \ln N_B} \right) \right) \dots (1)$$

TABLE III

Combined Data Representing Interdiffusion of Cr, Mo, W, and Tracer Diffusion of <sup>51</sup>Cr, <sup>99</sup>Mo, <sup>185</sup>W in Dilute Solution in Ferrite

|                | $D_0$ ,<br>mm <sup>2</sup> /s | $Q$ ,<br>kJ/mol | Combined Fit of Data                        |
|----------------|-------------------------------|-----------------|---------------------------------------------|
| Cr             | $240 \pm 14$<br>– 13          | $239.8 \pm 4.2$ | Present investigation; Refs. 7, 8, 9, 15–18 |
| Mo             | $78.5 \pm 6$                  | $225.5 \pm 4.6$ | Present investigation; Refs. 10, 11, 19, 20 |
| W <sub>1</sub> | $157 \pm 18$<br>– 17          | $243.5 \pm 7.1$ | Present investigation; Ref. 20              |
| W <sub>2</sub> | $262 \pm 46$<br>– 40          | $237 \pm 25$    | Refs. 8, 21, 22                             |

where  $D_A^*$ ,  $D_B^*$  are the tracer diffusion coefficients of A, B in the alloy AB.  $N_A$ ,  $N_B$  are the mol fractions of A and B and  $\gamma_B$  is the activity coefficient of B. For a very dilute solution of A in B, we can write  $N_A \ll N_B$ , hence

$$\bar{D}_{AB} \simeq D_A^* N_B \left( 1 + \frac{\partial \ln \gamma_B}{\partial \ln N_B} \right) \dots (2)$$

As  $N_A \rightarrow 0$ ,  $N_B \rightarrow 1$ , and  $\ln \gamma_B \rightarrow 1$ , hence  $\bar{D}_{AB} \simeq D_A^*$  for  $N_A \rightarrow 0$ . Verification of this equality has been reported<sup>14</sup> for the iron-vanadium system for up to 10 at.-% V.

In the present investigation steps were taken to ensure the availability of a supply of alloys containing a very low concentration of solute so that if  $\bar{D}$  was found to vary strongly with composition, a good extrapolation to  $N_A \rightarrow 0$  could be obtained. In fact, no concentration-dependence was found and so the extrapolation is thought to be even more reliable.

### Tracer Diffusion in Ferrite

Tracer diffusion of <sup>51</sup>Cr in ferrite has been studied by several workers<sup>8,15–18</sup>; the reported measurements are shown in Fig. 2. It can be seen that there is very good agreement between interdiffusion and tracer diffusion coefficients. Tracer diffusion of <sup>99</sup>Mo in ferrite has been studied and the available measurements<sup>19,20</sup> are shown in Fig. 3. Again there is good agreement between the reported values. Tracer diffusion of <sup>185</sup>W in ferrite is shown in Fig. 4, where the results<sup>8,20–22</sup> fall into two groups. The experimental scatter for each group does not explain the difference between the two. The results of Kucera<sup>20</sup> *et al.* agree quite closely with those of the present investigation but differ from the results of the second group<sup>8,21,22</sup> shown in Fig. 4. Unfortunately, owing to a lack of information<sup>20–22</sup> concerning the details of the measurements, it is not possible to decide which are the more reliable.

In view of the apparent lack of significant difference between tracer and interdiffusion measurements, selected data in the literature for diffusion in ferrite in each of the three systems have been taken with data of the present investigation (see Fig. 2) and treated statistically, fitting a straight line to  $\ln D$  vs.  $1/T$  by the method of least squares to give the recommended values shown in Table III. For the FeW system the values from the more comprehensive results of the present

TABLE IV

Comparison of Combined Interdiffusion and Combined Tracer Diffusion Measurements in Dilute Solution in Austenite

|                                         | $D_0$<br>mm <sup>2</sup> /s | $Q$ ,<br>kJ/mol | Combined Fit<br>of Data                   |
|-----------------------------------------|-----------------------------|-----------------|-------------------------------------------|
| <sup>51</sup> Cr, Tracer                | 4020 + 880<br>- 780         | 314 ± 28        | Refs. 8, 15, 17                           |
| 0-28.3 at.-%<br>Cr, Inter-<br>diffusion | 6.27 + 0.46<br>- 0.42       | 252.3 ± 18.4    | Present<br>investigation;<br>Refs. 7, 9   |
| <sup>99</sup> Mo, Tracer                | No Data Available           |                 |                                           |
| 0-5.05 at.-%<br>Mo, Inter-<br>diffusion | 3.60 + 0.40<br>- 0.36       | 239.8 ± 26.4    | Present<br>investigation,<br>Refs. 10, 12 |
| <sup>185</sup> W, Tracer                | 4205 + 8738<br>- 2839       | 317 ± 28        | Refs. 8, 22                               |
| 0-29 at.-%<br>W, Inter-<br>diffusion    | 13 + 16<br>- 17             | 267.4 ± 10.8    | Present<br>investigation                  |

work have been combined with those of Kucera<sup>20</sup> to give a recommended value, while the measurements due to the second group of workers<sup>8,21,22</sup> have been treated separately and are also given in Table III.

#### Tracer Diffusion in Austenite

Tracer diffusion in austenite below ~ 1050° C may be subject to a grain-boundary effect unless special precautions are taken.<sup>8,17</sup> Only those measurements regarded as sufficiently careful below 1050° C are shown in Figs. 2-4.

Tracer diffusion measurements<sup>8,15,17</sup> shown in Fig. 2 for <sup>51</sup>Cr in austenite are in reasonable agreement. Similarly, the measurements<sup>8,22</sup> for <sup>185</sup>W in austenite agree closely. No measurements of tracer diffusion of <sup>99</sup>Mo in austenite could be found in the literature. It can be seen from Figs. 2-4 that measurements for tracer diffusion and interdiffusion differ considerably. Values of each type of coefficient have been grouped separately and a least-squares analysis used to produce the values shown in Table IV.

There appears to be a large discrepancy between the reported values for tracer diffusion and interdiffusion in austenite. Since  $\bar{D}_{AB} \approx D_A^*$  as  $N_A \rightarrow 0$  (from Equation (2)), a difference between interdiffusion and tracer diffusion measurements is not expected. Furthermore, for the iron-chromium system the relationship between  $\bar{D}_{AB}$  and  $D_A^*$  may be derived without resorting to this approximation.

From equation (2) we have  $\bar{D}_{FeCr} = D_{Cr}^* N_{Fe} \left( 1 + \frac{\partial \ln \gamma_{Fe}}{\partial \ln N_{Fe}} \right)$ . It has been shown<sup>13</sup> that for a binary system such as FeCr

$$\frac{\partial \ln \gamma_{Fe}}{\partial \ln N_{Fe}} = \frac{\partial \ln \gamma_{Cr}}{\partial \ln N_{Cr}}$$

The variation of  $\gamma_{Cr}$  with  $N_{Cr}$  for the austenite phase in the FeCr system has been measured by several workers.<sup>1,23,24</sup> Moore<sup>1</sup> has reviewed the available data and from a statistical treatment of these data it may be shown that between 1 and 14 at.-% Cr

$$\frac{\partial \ln \gamma_{Cr}}{\partial \ln N_{Cr}} = 0.08$$

For temperatures between 1050 and 1390° C no consistent variation of  $\gamma_{Cr}$  with temperature was observed. This implies that  $\bar{D}_{FeCr} = D_{Cr}^* N_{Fe} 1.08$ , and since  $N_{Fe} \approx 1$  for a dilute solution of Cr in Fe there is not expected to be any measurable difference between  $\bar{D}_{FeCr}$  and  $D_{Cr}^*$  within the limits of resolution of the available techniques. However, the observed difference (Fig. 2) approaches an order of magnitude at 1300° C. It was not possible to repeat this calculation for Fe-Mo and Fe-W, owing to the lack of available thermodynamic data.

A possible reason for the difference between measurements is the presence of a systematic error in the measurements of tracer diffusion. Measurement of a tracer diffusion coefficient often involves plating a thin layer of the radioactive tracer on to a substrate and a subsequent evaluation of the redistribution of the tracer in the substrate is used to determine the tracer diffusion coefficient. The corresponding 'thin-film' solution to Fick's second equation (for constant  $D$ )<sup>25</sup> predicts the tracer distribution after time  $t$  but involves the approximation that the thin tracer layer (thickness  $h$ ) goes into solution in the substrate in a time that is short compared with  $t$ . Lundy and Murdock<sup>26</sup> have studied the error involved in making this approximation and concluded that for the case where there is complete solubility between substrate and tracer, the condition  $h < 0.1 \sqrt{Dt}$ , will cause insignificant error. However, in many cases there is only limited solid solution of the tracer in the substrate, and tracer measurements of Cr, Mo, and W in fcc Fe fall into this category. In these cases taking the tracer into solution is a slower process controlled by the solubility ( $C_0$ ) of the tracer in the substrate in addition to the diffusion rate and the above criterion is not applicable. Kidson<sup>27</sup> has shown that if the terminal solubility is low then there will be an initial period  $0 < t < \tau$  during which the concentration of the solute at the surface is effectively held fixed at the concentration  $C_0$ . Malkovitch<sup>28</sup> has shown that the appropriate criterion determining whether or not a significant error will occur from a specimen of unit area is  $t \geq \pi d^2 / D \alpha^2$ , where  $t$  is the diffusion time,  $d$  is the film thickness,  $D$  the diffusion coefficient, and  $\alpha$  the atomic fraction of solute at the terminal solubility  $C_0$ . Bowen,<sup>15</sup> whose measurements of tracer diffusion of <sup>51</sup>Cr are representative of tracer measurements in austenite, used plated layers 0.5  $\mu$ m thick. At 1330° C, the parameter  $\pi d^2 / D \alpha^2$  indicates that a diffusion time  $\geq 115$  min is required and that the diffusion time employed by Bowen<sup>15</sup> (35 min) will result in an erroneous value of  $D_{Cr}$ . Bowen<sup>15</sup> also investigated plated layers estimated, using a weighing technique, as 0.1  $\mu$ m thick; for these the required diffusion time for an insignificant error is  $\geq 5$  min. However, the diffusion coefficients obtained from samples with either plating thickness are in very close agreement but still an order of magnitude greater than the corresponding interdiffusion measurements, so this simple explanation of the anomaly is inadequate. If, however, a mechanism of this type is responsible then it is necessary to postulate the existence of some other factor that may stabilize the bcc phase or otherwise retard the dissolution of the tracer in the underlying matrix. For example, the formation of an impurity phase in the tracer layer during electroplating or the formation of an oxide film that subsequently dissolves in the course of the heat-treatment would not be detected on many tracer measurements, but would lead to an erroneous value of  $D_{Cr}$ .

Linearity of the  $\log c$  vs.  $x^2$  plot, which is often used as a check on the correct application of this analysis, is not a very sensitive test for the type of error described above, and erroneous application of this analysis can be shown to lead to values of  $D$  that are too large. The tracer measurements of  $D_\gamma$  in the FeCr system are higher than corresponding interdiffusion measurements to an extent which cannot be explained on any fundamental basis, and despite the lack of reliable thermodynamic data for the FeW system a similar systematic error may be the reason why the tracer measurements of  $D_\gamma$  by Kieszniowski,<sup>22</sup> who used a plating thickness of 0.5  $\mu\text{m}$ , are substantially greater than corresponding interdiffusion measurements.

### Conclusions

(1) Interdiffusion in the systems FeCr, FeMo, and FeW may be represented over the given ranges of concentration by the data given in Table II.

(2) Over the composition and temperature ranges studied in the above systems, no evidence of any concentration-dependence of interdiffusion coefficients was found for either ferrite or austenite. This is consistent with interdiffusion measurements of other workers.

(3) *Ferrite*. For dilute solutions at the iron-rich end of the FeCr and FeMo systems the agreement between tracer and

interdiffusion measurements is good and values have been combined (Table III). For the FeW system, owing to the scatter in tracer diffusion measurements the agreement is not as good. The values recommended are a combination of the measurements of the present investigation and those of Kucera *et al.*<sup>20</sup>

(4) *Austenite*. For diffusion in austenite dilute solutions at the iron-rich end of the systems FeCr and FeW, the available data suggest that there is a significant difference between tracer and interdiffusion measurements. This unexpected difference cannot be explained on a fundamental basis and is inconsistent with the available thermodynamic data for the FeCr system. The reason for this discrepancy is not clear but may be that a systematic error arises in the tracer technique. The values that may be taken as representative of interdiffusion and tracer diffusion, respectively, in dilute solution are given in Table IV.

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